LEWIS ELECTRON DOT STRUCTURES 11

GN Lewis (1875–1945) was an American chemist and one of the first people to come up with a theory of electronic bonding. He used dots to represent electron structures.

Atoms transfer or share electrons to form bonds. These bonds hold the atoms together. If an atom loses an electron, it forms a positive ion. If it gains an electron it forms a negative ion.

1 What is the electrical charge on an electron?

Metals tend to lose electrons and non-metals tend to gain or share electrons in bonds. The transfer of electrons results in **ionic bonds**. Sharing electrons results in **covalent bonds**. There is a shorthand way of showing how electrons are involved in bonding: **'electron dot' formulae**. These show only the valence (outer) electrons, leaving out the inner shells of electrons. Therefore, there may be many more electrons present but the 'inner' levels are not illustrated.

2 When studying bonding, why can the inner shells of electrons be left out?

Figure 1 uses Lewis electron dot structures to show how sodium (Na, atomic number 11, electronic configuration: 2, 8, 1) and chlorine (Cl, atomic number 17, electronic configuration: 2, 8, 7) atoms react to form the ionic compound, sodium chloride (NaCl).

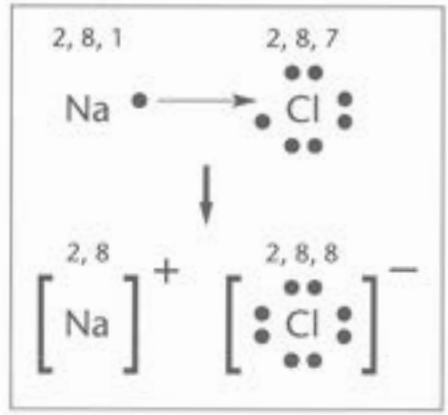


Figure 1

3 What is meant by *electronic configuration*?

4 Copy figure 1 and add labels to describe what occurs.

5 Give the symbol and electronic configuration of potassium (atomic number 19).

6 Give the symbol and electronic configuration of fluorine (atomic number 9).

7 Use Lewis electron dot structures to show how potassium and fluorine react to form potassium fluoride (KF).

8 What type of bonding is present in both sodium chloride and potassium fluoride?

9 Use Lewis electron dot structures to show how magnesium (atomic number 12) and oxygen (atomic number 8) react to form magnesium oxide (MgO).

Electron-dot formulae can also be used to show covalent bonds. Figure 2 shows how chlorine can form chlorine (Cl_2) molecules. Figure 3 shows how oxygen atoms can form an oxygen molecule (O_2).

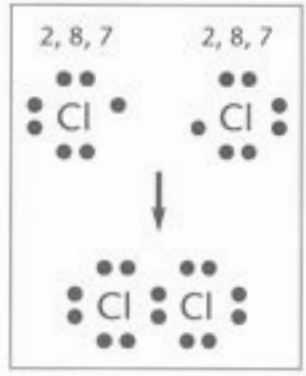


Figure 2

10 Copy figure 2 and add labels to explain what happens between the atoms.

11 Use Lewis electron dot structures to show how fluorine atoms react to form fluorine (F_2) molecules.

12 Copy figure 3 and add labels to explain what happens between the atoms.

13 Outline the ways electrons are involved in ionic bonding and compare this to the ways electrons are involved in covalent bonding.

Tetrachloromethane (CCl_4) is a covalent molecule. Carbon has an atomic number of 6. Figure 4 shows part of the structure of the tetrachloromethane molecule.

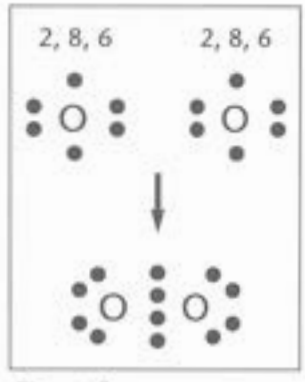


Figure 3

14 Copy figure 4 and show how the electrons are arranged in the molecule.

Research **15** GN Lewis also developed a theory of acids and bases. Outline this theory.

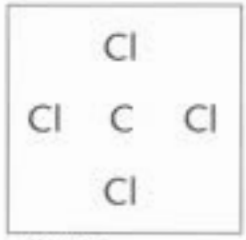


Figure 4

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